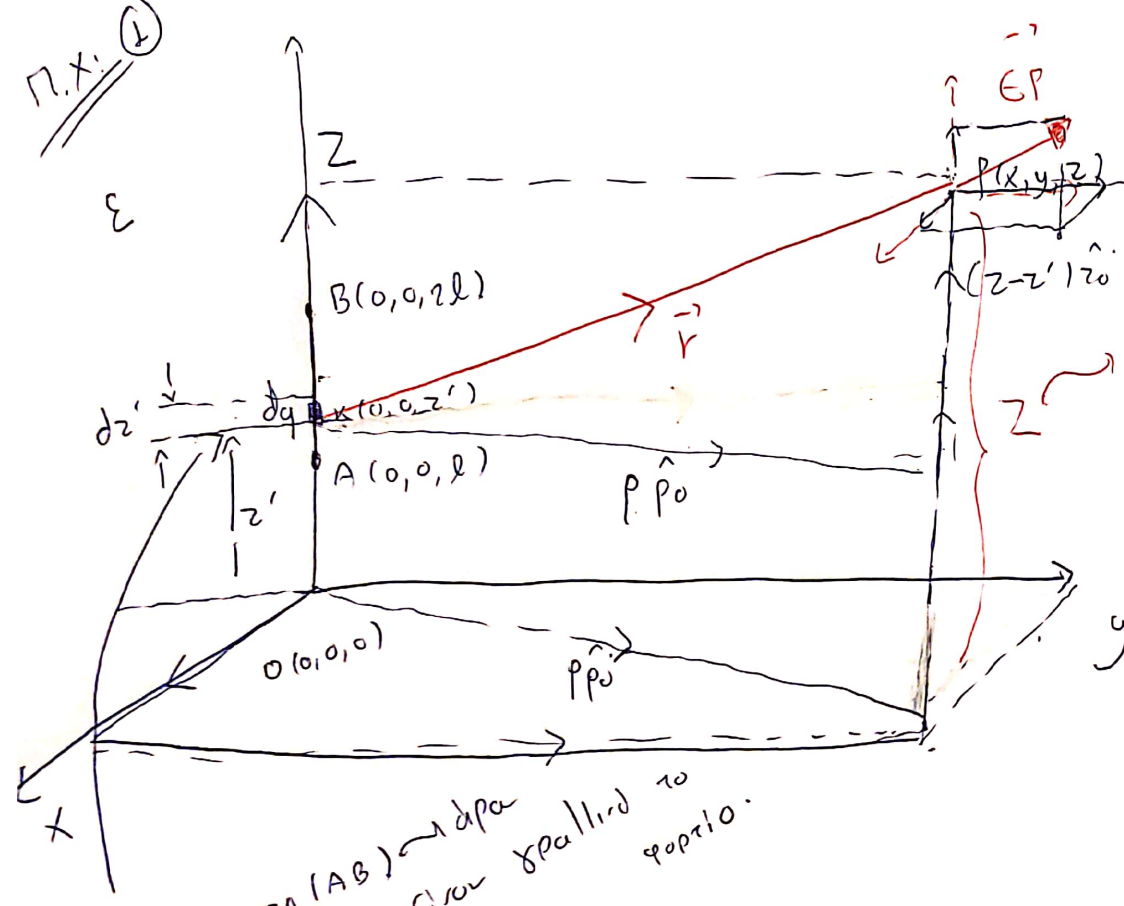




$f(Q, \text{O.D.}, \sigma_0(AB)) \rightarrow \text{dpa}$
 Error scaled to ratio.

$\rho \in \mathbb{R}^+$
 $dq = \rho \cdot dl = \rho \cdot dz' \sim \int_0^Q dq = \int_l^{2l} \rho \cdot dz' \sim Q = \rho \cdot [2l - l] \leadsto Q = \rho \cdot l \cdot (c) \leadsto \rho = \frac{Q}{l} \text{ (C/m)}$
 $(c) \quad (C/m) \cdot (m)$
 $\hat{x}_0 \perp \hat{y}_0 \perp \hat{z}_0$

$$\vec{r} = \vec{r}_P = (x, y, z) - (0, 0, z') = (x - 0, y - 0, z - z') = x\hat{x}_0 + y\hat{y}_0 + (z - z')\hat{z}_0$$

$$E_P, \phi_P = ?$$

$$d\phi_r = \frac{dq}{4\pi\epsilon r}$$

$$d\epsilon_p = \frac{|dq|}{4\pi r^3} \quad -$$

$$d\varphi_p = \frac{dq}{4\pi\epsilon r} = \frac{\rho dz'}{4\pi\epsilon (x^2 + y^2 + (z-z')^2)^{3/2}} \rightarrow$$

$$\rightarrow \varphi_P = \int_0^{\varphi_P} d\varphi_P = \frac{\rho \ell}{4\pi\epsilon} \int_{\ell}^{2\ell} \frac{1}{(x^2 + y^2 + (z - z')^2)^{1/2}} dz' \rightarrow$$

→ $\frac{1}{2} \times 100 = 50\%$

$$r = \sqrt{x^2 + y^2 + (z - z')^2}$$

$$\rightarrow \varphi_P = \frac{\rho l}{4\pi\epsilon} \int_{-l}^l \frac{-d(z-z')}{(x^2+y^2+(z-z')^2)^{3/2}} = \frac{\rho l}{4\pi\epsilon} \left[-\ln[(z-z') + \sqrt{x^2+y^2+(z-z')^2}] \right]_{-l}^l$$

$d z' = -dz + dz' = -dz$
 $= -[dz - dz'] =$
 $= -d[z - z']$

$\downarrow$ h' ε' r' u
 $z - z' = a$

$$\rightarrow \varphi(x, y, z) = \frac{\rho l}{4\pi\epsilon} \left(-\ln(z-l + \sqrt{x^2+y^2+(z-l)^2}) + \ln(z+l + \sqrt{x^2+y^2+(z+l)^2}) \right)$$

$\nearrow$ ε' d' τ' o x l' o' s r' i' m
 τ' o z - z'
 o' u' t' o' x² + y²

$$\sim \varphi(x, y, z) = \frac{\rho l}{4\pi\epsilon} \ln \left[\frac{(z-l) + \sqrt{x^2+y^2+(z-l)^2}}{(z+l) + \sqrt{x^2+y^2+(z+l)^2}} \right] = \varphi(x, y, z)$$

$\downarrow$ ε' o' p' p' z' o' m' l' a
 $\rho^2 = x^2 + y^2, z = z$

$$\hookrightarrow \varphi(r, z) = \frac{\rho l}{4\pi\epsilon} \ln \left(\frac{z-l + \sqrt{\rho^2 + (z-l)^2}}{z+l + \sqrt{\rho^2 + (z+l)^2}} \right)$$

$$\left(\ln[x + \sqrt{x^2 + a^2}] \right)' =$$

$$= \frac{1}{x + \sqrt{x^2 + a^2}} (x + \sqrt{x^2 + a^2})' =$$

$$= \frac{1}{x + \sqrt{x^2 + a^2}} \cdot \left(1 + \frac{x}{\sqrt{x^2 + a^2}} \right) =$$

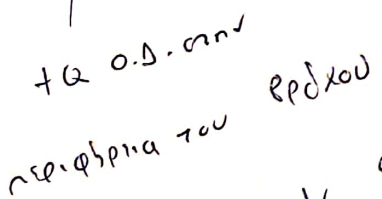
$$= \frac{1}{x + \sqrt{x^2 + a^2}} \left(\frac{\sqrt{x^2 + a^2} + x}{\sqrt{x^2 + a^2}} \right) =$$

$$= \frac{1}{\sqrt{x^2 + a^2}}$$

$$\bullet d\vec{E}_P = \frac{1 dq}{4\pi\epsilon r^2} \vec{r} = \frac{|\rho l dz'|}{4\pi\epsilon (x^2+y^2+(z-z')^2)^{3/2}} (x\hat{x} + y\hat{y} + (z-z')\hat{z})$$

$\downarrow$
 $\int$

П.Х: (2)



$$\varphi \varphi^2 \leadsto \int \vec{\mathcal{L}} = \int \rho \hat{\rho}_0 + \rho \int \varphi \hat{\varphi}_0 + \int z \hat{z}_0$$

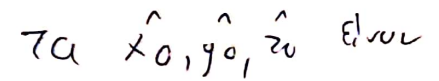
$$z=0 \text{ (stet.)}$$

$$\varphi \in L_0(\mathbb{R})$$

$$\Rightarrow x_2 = p l \cdot a \cdot 2n \sim$$

$$Q = p \lambda$$

$$\Rightarrow p \lambda = \frac{Q}{2\pi a} \quad (C/m)$$



ερωτα $\hat{n}_0, \hat{\varphi}_0, \hat{J}_0, \hat{\rho}_0 \Delta \in \mathcal{N}$

EINA1 मात्र

ως προς συνθήκη
(α ήχο είναι).

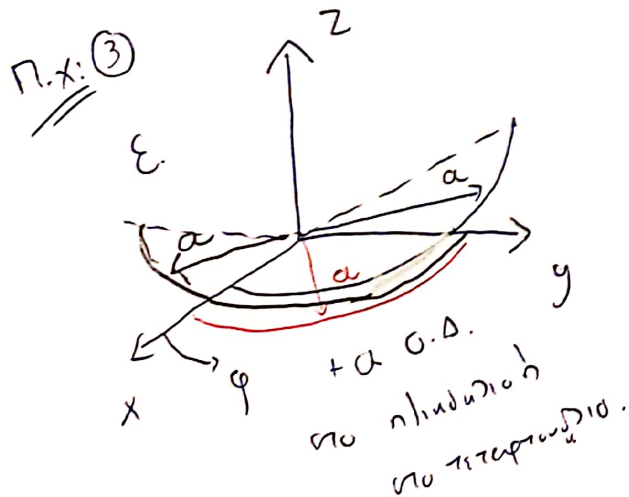
$$d\varphi_P = \frac{dq}{4\pi\epsilon r} = \frac{\rho l a d\varphi}{4\pi\epsilon (a^2+z^2)^{3/2}} \quad \leadsto \quad \varphi_P = \int_0^{\varphi_P} d\varphi_P = \frac{\rho l a}{4\pi\epsilon} \int_0^{\varphi_P} \frac{1}{(a^2+z^2)^{3/2}} d\varphi = \frac{\rho l a}{2\epsilon} \frac{1}{(a^2+z^2)^{1/2}} \quad (v)$$

$$\vec{dE}_P = \frac{|dq|}{4\pi\epsilon r^2} \vec{r} = \frac{\rho l a d\varphi}{4\pi\epsilon (a^2+z^2)^{3/2}} \left[a(-\hat{r}_0) + z\hat{z}_0 \right] = \frac{\rho l a d\varphi}{4\pi\epsilon (a^2+z^2)^{3/2}} (-a\cos\varphi\hat{x}_0 - a\sin\varphi\hat{y}_0 + z\hat{z}_0) \rightarrow$$

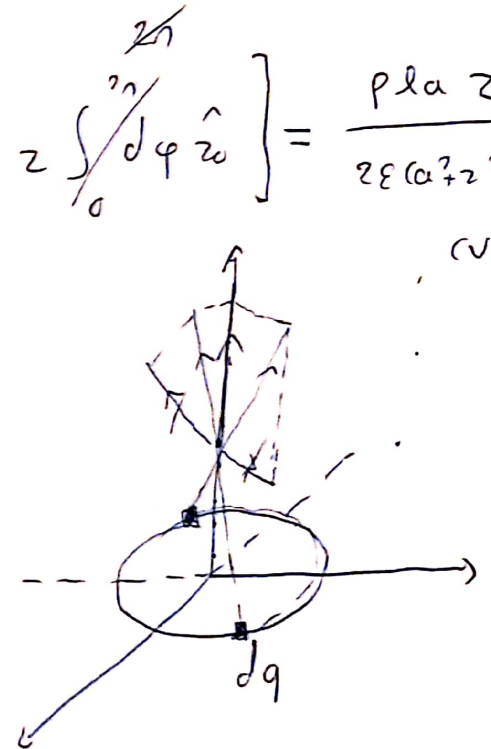
το $\hat{r}_0 \notin \mathcal{N}$
 είναι σταθερό ως
 προς φ , άρα κατά ω

$$\rightarrow \vec{E}_P = \int_0^{\varphi_P} \vec{dE}_P = \frac{\rho l a}{2} \left[-a \int_0^{\varphi_P} \cos\varphi d\varphi \hat{x}_0 - a \int_0^{\varphi_P} \sin\varphi d\varphi \hat{y}_0 + z \int_0^{\varphi_P} d\varphi \hat{z}_0 \right] = \frac{\rho l a z}{2\epsilon (a^2+z^2)^{3/2}} \hat{z}_0 \quad (v/m)$$

$$\rightarrow \vec{E}(0) = \vec{0}, \text{ γιατί } \varphi(0) = \frac{\rho l a}{2\epsilon} \frac{1}{(a^2)^{1/2}} = \frac{\rho l}{2\epsilon} \quad (v)$$



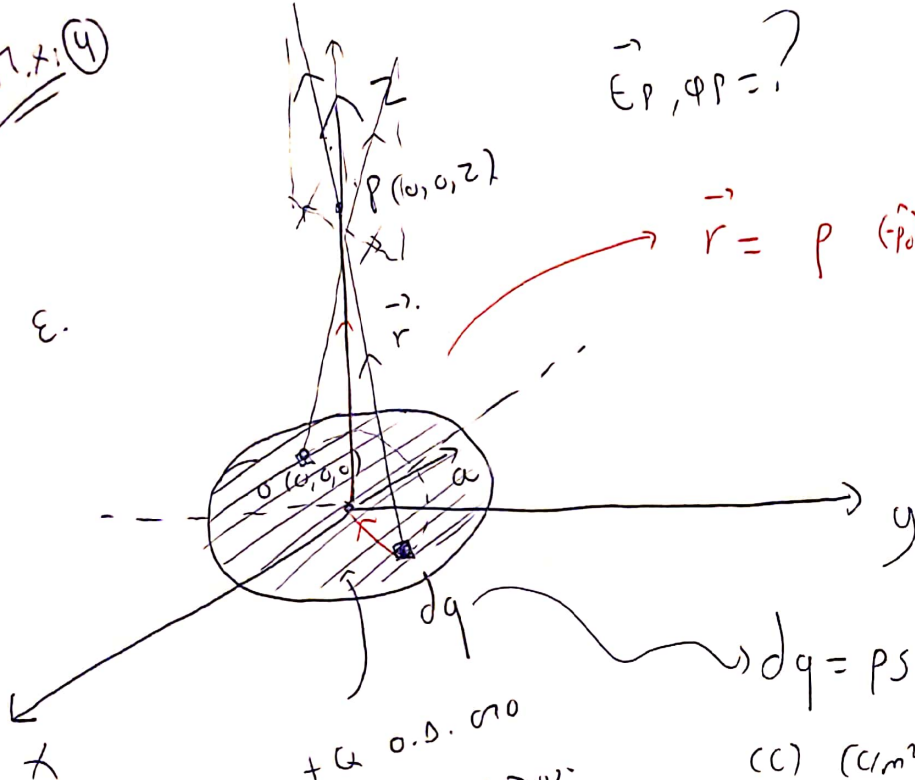
- $(\alpha) \varphi \in [0, \pi]$
 $(\beta) \varphi \in [0, \pi/2]$
 $(\gamma) \varphi \in [-\pi/2, \pi/2]$
- $\left. \begin{array}{l} \text{ } \\ \text{ } \end{array} \right\} \vec{E}_P, \varphi_P = ?$



Π. 4. (4)

$$\vec{E}(\rho, \varphi, z) = ?$$

ε.

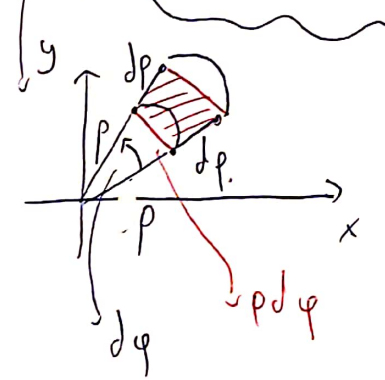


+ Q o.d. σ₀
ε (C/m²) (m²)

$$\vec{r} = \rho (\hat{\rho}) + z (\hat{z}) \quad \hat{\rho} \perp \hat{z}$$

$$r = \sqrt{\rho^2 + z^2} = (\rho^2 + z^2)^{1/2}$$

$$dS_{\rho\varphi z} = \begin{cases} \rho d\rho d\varphi \hat{z} \\ \rho d\varphi dz \hat{\rho} \\ d\rho dz \varphi \hat{\phi} \end{cases}$$



$$dS = (\rho d\varphi) d\rho = \rho d\rho d\varphi$$

$$dq = \rho_s dS \stackrel{z=0}{dz=0} = \rho_s \rho d\rho d\varphi$$

(C) (C/m²) (m²)

$$\int_0^{2\pi} dq = \rho_s \int_0^{2\pi} \int_0^a \rho d\rho d\varphi = \rho_s [\varphi]_0^{2\pi} \left[\frac{\rho^2}{2} \right]_0^a =$$

$$\frac{dq}{4\pi\epsilon r} = \frac{\rho_s \rho d\rho d\varphi}{4\pi\epsilon (\rho^2 + z^2)^{1/2}} \rightarrow \int_0^\varphi d\varphi = \frac{\rho_s}{4\pi\epsilon} \int_0^{2\pi} \int_0^a \frac{\rho d\rho d\varphi}{(\rho^2 + z^2)^{1/2}} \rightarrow$$

$$\rightarrow \varphi = \frac{\rho_s}{4\pi\epsilon} \int_0^{2\pi} \frac{d\varphi}{2\pi} \int_0^a \frac{\rho}{\sqrt{\rho^2 + z^2}} d\rho = \frac{\rho_s}{2\epsilon} \left[\sqrt{\rho^2 + z^2} \right]_0^a = \frac{\rho_s}{2\epsilon} \left[\sqrt{a^2 + z^2} - \sqrt{z^2} \right] =$$

$$|z| = \begin{cases} z, z \geq 0 \\ -z, z < 0 \end{cases} = \frac{\rho_s}{2\epsilon} \left[\sqrt{a^2 + z^2} - |z| \right], (V)$$

$$d\vec{E}_P = \frac{|dq|}{4\pi\epsilon r^3} \vec{r} = \frac{ps \, p \, dp \, d\varphi}{4\pi\epsilon (p^2+z^2)^{3/2}} \left[\underbrace{p(-\hat{p}_0) + z\hat{z}_0}_{p(-\cos\varphi\hat{x}_0 - \sin\varphi\hat{y}_0)} \right] = \frac{ps}{4\pi\epsilon} \left[\frac{-p^2\cos\varphi}{(p^2+z^2)^{3/2}} dp \, d\varphi \, \hat{x}_0 - \frac{p^2\sin\varphi}{(p^2+z^2)^{3/2}} dp \, d\varphi \, \hat{y}_0 + \right.$$

$$\left. + \frac{z \, p}{(p^2+z^2)^{3/2}} dp \, d\varphi \, \hat{z}_0 \right] \sim$$

$$\vec{E}_P = \int_0^{2\pi} \int_0^a d\vec{E}_P = \frac{ps}{4\pi\epsilon} \left[- \int_0^{2\pi} \cos\varphi \, d\varphi \int_0^a \frac{p^2}{(p^2+z^2)^{3/2}} dp \, \hat{x}_0 - \int_0^{2\pi} \sin\varphi \, d\varphi \int_0^a \frac{p^2}{(p^2+z^2)^{3/2}} dp \, \hat{y}_0 + \right.$$

$$\left. + z \int_0^{2\pi} d\varphi \int_0^a \frac{p}{(p^2+z^2)^{3/2}} dp \, \hat{z}_0 \right] = \frac{ps \, 2\pi \, z}{24\pi\epsilon} \int_0^a \frac{p}{(p^2+z^2)^{3/2}} dp \, \hat{z}_0 =$$

$$= \frac{ps \, z}{2\epsilon} \left[- \frac{1}{(p^2+z^2)^{1/2}} \right]_0^a \hat{z}_0 = \frac{ps \, z}{2\epsilon} \left(- \frac{1}{\sqrt{a^2+z^2}} + \frac{1}{\sqrt{z^2}} \right) \hat{z}_0 =$$

$$= \frac{ps \, z}{2\epsilon} \left[\frac{1}{|z|} - \frac{1}{\sqrt{a^2+z^2}} \right] \hat{z}_0 \quad (V/m)$$

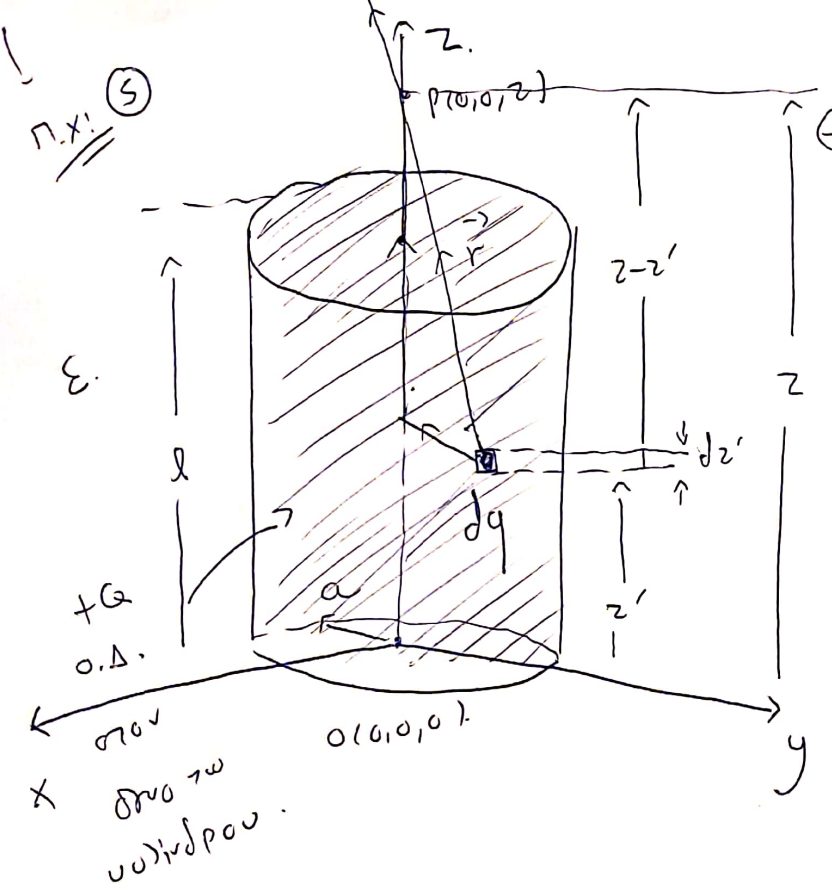
Όταν $a \rightarrow \infty$
(ο δίσκος προσλαμβάνει
άπειρο φορτίο)

$$\vec{E}_P = \frac{ps \, z}{2\epsilon} \left[\frac{1}{|z|} - 0 \right] \hat{z}_0 = \frac{ps \, z}{2\epsilon |z|} \hat{z}_0 =$$

$$\left(\frac{1}{(x^2+a^2)^{1/2}} \right)' = \left((x^2+a^2)^{-1/2} \right)' = -\frac{1}{2} (x^2+a^2)^{-3/2} \cdot 2x = -\frac{x}{(x^2+a^2)^{3/2}}$$

$$= \frac{x}{(x^2+a^2)^{3/2}}$$

$$\left. \begin{matrix} \frac{ps}{2\epsilon} (+\hat{z}_0), z > 0 \\ \frac{ps}{2\epsilon} (-\hat{z}_0), z < 0 \end{matrix} \right\} (V/m)$$



$\vec{r} = \rho(-\hat{\rho}_0) + (z-z')\hat{z}_0 \xrightarrow{\hat{\rho}_0 \perp \hat{z}_0} r = \sqrt{\rho^2 + (z-z')^2}$
 $\vec{r} = \rho(\cos\phi\hat{x}_0 - \sin\phi\hat{y}_0) + (z-z')\hat{z}_0 = (\rho^2 + (z-z')^2)^{1/2}$

$dq = \rho_v dV = \rho_v \rho d\rho d\phi dz' = \rho_v \rho d\rho d\phi dz' \sim$
 $(C/m^3)(m^3)$
 $\sim \int_0^Q dq = \rho_v \int_0^a \frac{a^2}{\rho} d\rho \int_0^{2\pi} d\phi \int_0^l dz = \rho_v \pi \frac{a^2}{2} l = \rho_v \pi a^2 l \sim$
 $\sim Q = \rho_v \pi a^2 l \sim \rho_v = \frac{Q}{\pi a^2 l} (C/m^3)$

$d\phi_p = \frac{dq}{4\pi\epsilon r} = \frac{\rho_v \rho d\rho d\phi dz'}{4\pi\epsilon (\rho^2 + (z-z')^2)^{1/2}} \longrightarrow$

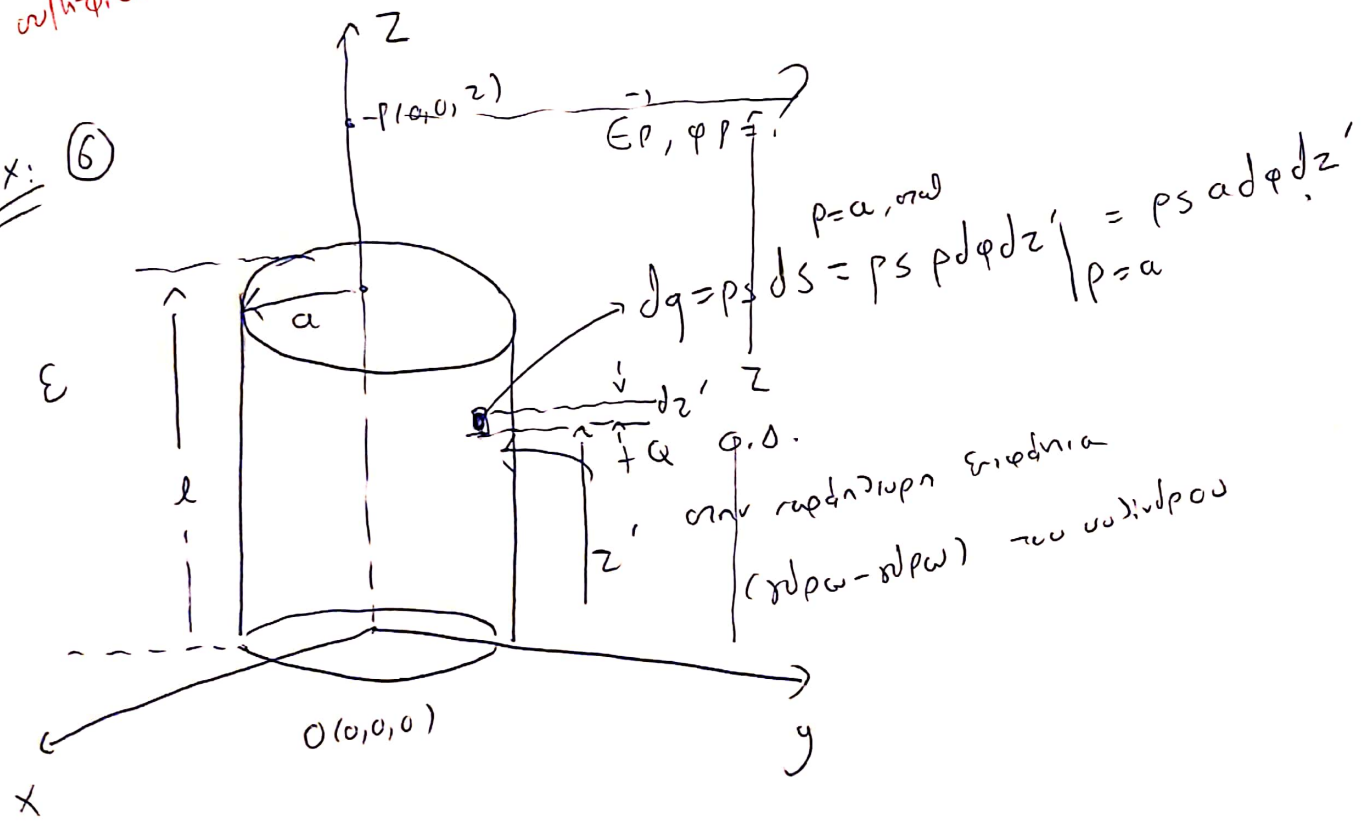
$\longrightarrow \phi = \frac{\rho_v}{4\pi\epsilon} \int_0^{2\pi} d\phi \int_0^l \int_0^a \frac{\rho d\rho dz'}{\sqrt{(\rho^2 + (z-z')^2)}} = \frac{\rho_v}{2\epsilon} \int_0^l \left[\sqrt{\rho^2 + (z-z')^2} \right]_0^a dz' =$
 $= \frac{\rho_v}{2\epsilon} \int_0^l \left[\sqrt{a^2 + (z-z')^2} - \sqrt{(z-z')^2} \right] dz' = \frac{\rho_v}{2\epsilon} \int_0^l \left(\sqrt{a^2 + (z-z')^2} - |z-z'| \right) dz' =$
 $= \dots (z-z') \delta r_0$

$$d\vec{E} = \frac{1}{4\pi\epsilon r^3} \vec{r} \quad \sim \quad \frac{\rho v \rho d\rho d\varphi dz'}{4\pi\epsilon (r^2 + (z-z')^2)^{3/2}} (-\rho \cos\varphi \hat{x}_0 - \rho \sin\varphi \hat{y}_0 + (z-z') \hat{z}_0) \quad \sim$$

$\hat{x}_0, \hat{y}_0$
 ↓ ρ cos φ $\hat{x}_0$
 ρ sin φ $\hat{y}_0$

$$\vec{E} = \frac{\rho v}{4\pi\epsilon} \int_0^l \int_0^{2\pi} \int_0^a \frac{\rho (z-z') d\rho d\varphi dz'}{(r^2 + (z-z')^2)^{3/2}} \hat{z}_0 \quad \sim \quad \vec{E} = \frac{\rho v}{2\epsilon} \int_0^l \int_0^a \frac{\rho (z-z') d\rho dz'}{(r^2 + (z-z')^2)^{3/2}} \hat{z}_0 = \dots (v/m)$$

Π.Χ: ⑥



hint
 να με σφο υαυαυαυα
 Ιαυα Ιαυα, να
 φοιταυαυα στα υαυαυα